

Miller Creek Collaring

Initial uncalibrated NDVI and shadow assessment

Survey flown 2025-10-21 | 132 surveyed points

Source imagery

DroneDeploy orthomosaics, 50 m AGL, GCP-controlled, EPSG:6514
Multispectral: 5 bands + alpha, 2.66 cm GSD (R, G, NIR, RE, unused, alpha)
Visible: RGB + alpha, 1.28 cm GSD
Survey points: Emlid RS3 RTK, all fixed solutions, EPSG:4326
GPS RMS below is the Lateral (horizontal) RMS from the Emlid export;
tilt is the pole tilt angle. Absent values are marked !!!NONE!!!

Method

$NDVI = (NIR - Red) / (NIR + Red)$, band 3 and band 1 of the MS ortho
Averaged per pixel over circular buffers of 0.10 m and 0.25 m radius
(44 and ~277 contributing pixels respectively; alpha = 0 excluded)
Shadow index after Rikimaru et al. (2002): $SI = [(1-R)(1-G)(1-B)]^{(1/3)}$
on 0-1 scaled visible bands, averaged over a 3 m radius buffer.
Higher SI means darker / more shadowed.

Caveat

The MS ortho carries isCalibrated = False and 8-bit bands, so NDVI here
is a relative index on raw DN, not reflectance NDVI. Values are
comparable within this survey only, not across dates or sensors.

Summary statistics

field	n	min	mean	median	max
ndvi_0o1	132	-0.317	+0.301	+0.308	+0.631
ndvi_0o25	132	-0.338	+0.292	+0.302	+0.650
shadow_idx	132	+0.466	+0.738	+0.755	+0.899
gps_rms_m	132	0.015	0.029	0.021	0.665
tilt_deg	127	0.5	14.4	7.5	78.5

5 of 132 points have no tilt_deg recorded; marked !!!NONE!!! on the chips.











































